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Artificial maturation and larval production of the limpet *Patella aspera* Röding, 1798 (Patellogastropoda, Mollusca): Enhancing fertilization success of oocytes using NaOH-alkalinized seawater

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Abstract

Patella aspera is a valuable limpet species native from the Macaronesian Region, which commercial exploitation is regulated due to overfishing. Aquaculture production of the species has the potential to be a sustainable alternative to the exploitation of this natural resource. This is the first study of the larval production of *P. aspera* with oocytes obtained from dissected gonads and artificially matured using NaOH-alkalinized seawater. An artificial maturation experiment obtained the higher ratio of matured oocytes employing alkalinized seawater at pH 9.0 and 9.5 from 3 to 5 h. In vitro experiments combined similar artificial maturation conditions: pH 9 and 9.5 (plus pH 8 as control) during 1:30, 3:00 and 4:00 h; with three fertilization times (1:30, 3:00 and 24:00 h) to determine the combination of factors leading to higher larval production. Results show increased larval production, ranging from 50% to 75% total fertilized oocytes, using alkalinized baths at pH 9 from 3 to 4 h, followed by 24 h fertilization time at 19 ± 1°C. The present methodology ensures the production of *P. aspera* larvae, a mandatory step for further studies focused on the larviculture and aquaculture production of this limpet species.

KEYWORDS

activation, fertilization, gametes, limpets, maturation, pH

1 | INTRODUCTION

Patella aspera Röding, 1798 is a true limpet species (Patellogastropoda) native from the Macaronesian Region (Sousa et al., 2017) and an important gastronomic and economical resource for the regional economy (Fernandes et al., 2019; Sousa, Vasconcelos, Henriques, et al., 2019). Currently, the limpet populations are compromised by the overexploitation (González-Lorenzo et al., 2015; Núñez et al., 2003), and legal regulations and management measures were imposed to reduce the anthropogenic impact and restore the native stocks in Madeira (Sousa, Vasconcelos, Henriques, et al., 2019; Sousa, Vasconcelos, Riera, et al., 2019), Azores (Serrão et al., 2010) and the Canary archipelagos (DECRETO 182/2004; Núñez et al., 2003).

Limpets represent an opportunity to diversify the local production in aquaculture in the Macaronesian Region, a strategy that could promote the consolidation of the industry through a sustainable development (Schmidt et al., 2011). Controlled production of the species in aquaculture facilities provides a sustainable alternative to the exploitation of the natural resources and could be applied for restocking purposes. The aquaculture of limpets is a recent research field, so few bibliographical and methodological resources are available (Ferranti et al., 2018; Guallart, 2010; Guallart et al., 2020;

Mau & Jha, 2018a), and zootechnical requirements are mostly unknown (Mau & Jha, 2018a). Most efforts on limpets aquaculture research have been focused on the acclimation of adult stocks (Hua & Ako, 2020), design of artificial diets (Hua & Ako, 2016; Mau & Jha, 2018b), reproduction in captivity (Aquino De Souza et al., 2009; Ferranti et al., 2018; Hodgson et al., 2007; Hua & Ako, 2020; Kay & Emlet, 2002; Mau et al., 2018), larviculture and settlement induction (Ferranti et al., 2018; Hua & Ako, 2020).

Larval production is a mandatory step for further studies focused on the larviculture and aquaculture production of limpets (Guallart et al., 2013). In captivity, limpet reproduction follows two different strategies. One method relies in the spawning induction, in which mature specimens are submitted to certain physical or chemical conditions to induce spawning of the gametes (Ferranti et al., 2018; Guallart, 2010; Hua & Ako, 2020; Kay & Emlet, 2002). This method has some disadvantages, such as requiring acclimated individuals (Ferranti et al., 2018; Hua & Ako, 2020) and uncertain spawning success (Ferranti et al., 2018; Guallart, 2010; Kay & Emlet, 2002). Nevertheless, relative success on the spawning induction of limpets of the genus *Cellana* was found when hormonal treatments were employed (Hua & Ako, 2020).

The alternative method implies the dissection of mature specimens to extract the gonads and release the gametes (Guallart, 2010; Guallart et al., 2020). However, only 5% of these oocytes could be suitable for fertilization (Dodd, 1957). The fertilization success can be enhanced employing a treatment called activation which consists in using an alkalinized seawater bath that induces the artificial maturation of the oocytes (Corpuz, 1981; Guallart, 2010; Pérez et al., 2016; Pérez et al., 2019; Souza & Silva, 2009). NaOH and NH_4OH were the most common alkalinizing agents used for the activation in several limpet species (see Table 1 for details). The success of the activation is determined by the pH and duration of the bath (Aquino De Souza et al., 2009; González-Novoa, 2014; Guallart et al., 2020; Pérez et al., 2016; Pérez et al., 2019). Regarding the sperm activation, in limpets it only requires to dilute the sperm in seawater during a given time (Hodgson et al., 2007; Pabst-Fernández, 2015; Pérez et al., 2019). This methodology does not require a previously acclimated reproductive stock (Aquino De Souza et al., 2009; Guallart, 2010; Hodgson et al., 2007).

The aim of the present study is to develop a consistent protocol for the reliable production of *P. aspera* larvae. The strategy involves the gonad extraction to release the gametes and NaOH-alkalinized seawater baths to induce the maturation of the oocytes. Therefore, the influence of the pH, activation time and fertilization time over the larval production were evaluated in the current study.

2 | MATERIAL AND METHODS

A total of 88 adult specimens of *P. aspera* were collected from floating fish cages and rocks on the southern coast of Madeira Island (Portugal; Figure 1) during winter time (between December 2019 and February 2020) to obtain higher proportions of mature specimens (Sousa et al., 2017). After capture, the specimens were immediately transported to aquaculture facilities at Centro de Maricultura da Calheta, measured (carapace length, CL; Digital Stainless Calliper, precision = 0.01 mm), weighted (total weight, TW; Precision Balance RADWAG WTC 600.1, range = 600 g, precision = 0. 01 g, Poland), tagged with vinyl sheets attached with cyanoacrylate and transferred to 200 L tanks. The adults were maintained in an open system

TABLE 1 Conditions of the alkalinized seawater baths for the artificial maturation of limpet oocytes (Patellogastropoda)

Species	Base	pH	AT	References
Lottia gigantea Gray in G. B. Sowerby I, 1834	NH_4OH	8.5–9.0	10–38 min	Gould et al. (2001)
Patella aspera Röding, 1798	NaOH	9.0	3 h	Present study
Patella caerulea Linnaeus, 1758	NH_4OH	8.9	1–7 min	Wanninger et al. (1999)
Patella depressa Pennant, 1777	NaOH	9.5	1–5.5 h	Aquino De Souza et al. (2009)
	NH_4OH	9.0	10 min	Pérez et al. (2019)
Patella ferruginea Gmelin, 1791	NaOH	9.0	1–2 h	Guallart (2010); Guallart et al. (2020)
Patella ulyssiponensis Gmelin, 1791	NaOH	9.0	4–5 h	Pérez et al. (2019)
	NH_4OH	9.0	10 min	Hodgson et al. (2007); Pérez et al. (2019)
Patella vulgata Linnaeus, 1758	NaOH	8.9	5–7 min	Damen and Dictus (1994)
	NH_4OH	9.0	10–20 min	Pérez et al. (2019)
	NH_4OH	9.5	10 min	Pérez et al. (2019)

Note: The species, base employed, pH and activation time (AT) are indicated.

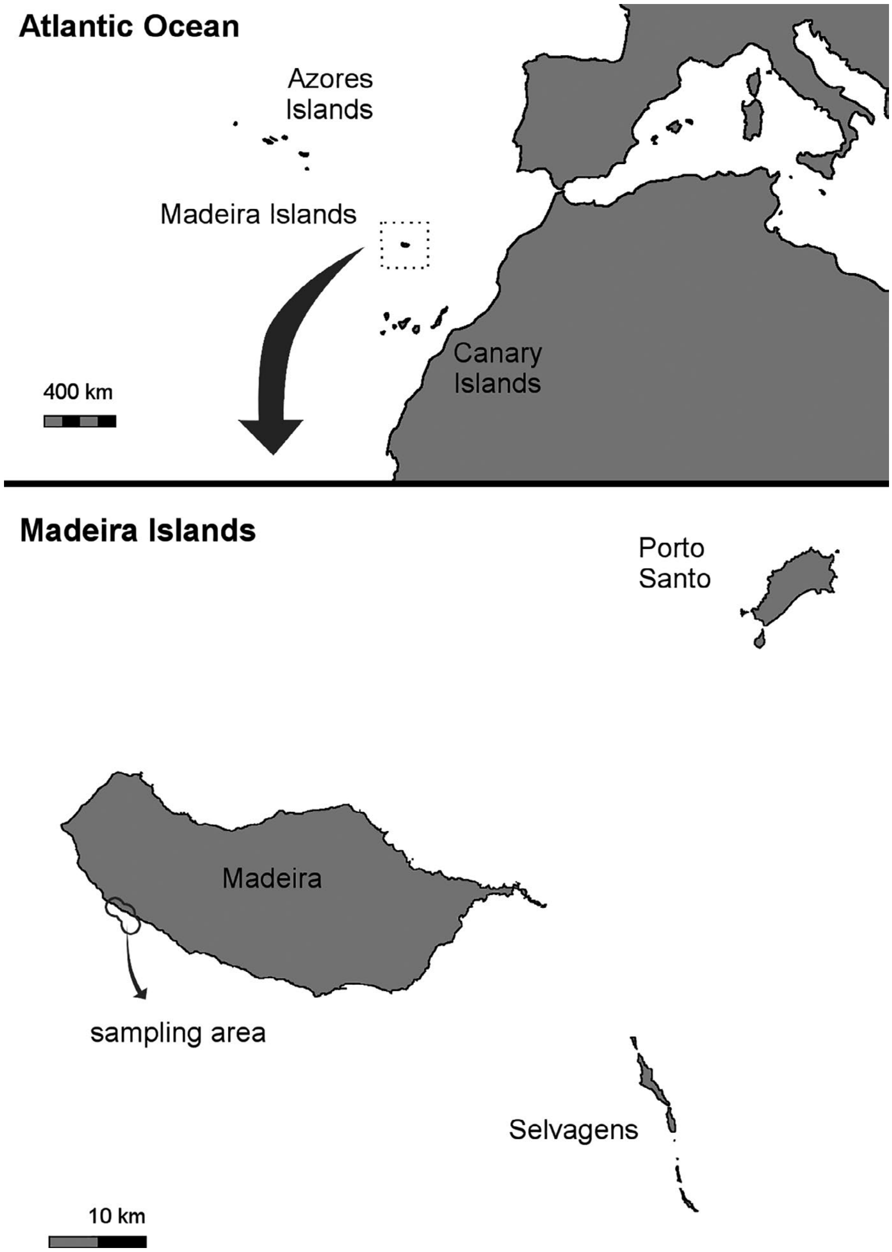

FIGURE 1 *Patella aspera*. Geographic map of Madeira (Portugal) showing the sampling area

(1.85 L/min), with filtered seawater (50 μm), and constant aeration at 20 ± 1°C (Oxygen Probe, Handy Polaris v.3.11., OxyGuard®) and 37 ± 1 g/L (Hand-Held Refractometer A.S.D. 0–100, ATAGO). The experiments were carried out between 4 and 10 days from the arrival of the specimens to avoid influencing the gonad quality.

2.1 | Extraction of the gametes

Before each experiment, the sex was determined by biopsy using 1 ml syringe. The dissection was performed cutting the mantle on the left side of the animal using a scalpel, and then, the foot was upturned to expose the gonads. The female gonads were weighted (gonad weight, GW; Precision Balance XT220A, range = 0.01–220 g, precision = 0.0001 g, Switzerland) and gonadosomatic index (GSI) was calculated as $GSI = GW \times FW^{-1} \times 100$, being GW the gonad weight and FW the fresh weight (FW = total weight - shell weight), following Sousa et al. (2017). The female gonads were then gently stirred in 100 ml FSU (filtered seawater treated with UV) using a Pasteur's pipette to release the oocytes. The oocyte suspensions were cleaned using a double sieve system: the first sieve, a 200 μm mesh was used to remove debris pieces allowing the passage of the oocytes to the second sieve, a 55 or 100 μm mesh used to collect and wash repeatedly the oocytes using FSU. Once washed, the oocytes of four females were pooled in 200 ml FSU. Regarding the sperm extraction, the male gonads were exposed, and sperm was collected using a Pasteur's pipette. This method was effective collecting 'mostly clean' sperm but did not allow to calculate the male's GSI. The sperm of four males was pooled in 100 ml of FSU.

2.2 | Artificial maturation

Four females (CL = 43.9 ± 1.4 mm, TW = 11.0 ± 0.9 g; GSI = 28 ± 10) collected in December 2019 were selected for this experiment. The FSU was alkalinized using NaOH to prepare five pH treatments: 8.5, 9.0, 9.5, 10.0 and 10.2 ± 0.1 (pH Meter GLP 2, CRISON). Non-treated FSU was used as control treatment (pH 8.0 ± 0.1). The oocytes were incubated at seven different activation times (AT): 0:30, 1:00, 1:30, 2:00, 3:00, 4:00 and 5:00 h. Therefore, a total of 42 combined treatments resulted from the combination of pH × AT, with three replicates *per* combined treatment. Each replicate consisted in a single plastic cup (100 ml) filled with 20 ± 1 ml of pH treatment. The experiment started when 1 ml of oocytes was added in all the replicates. The average density was 74 ± 12 oocytes ml^{-1}.

Replicates were numbered and distributed in a grid marked with coordinates following a random pattern (https://stattrek.com). At the end of each AT, the corresponding replicates were analysed at real time by two observers (nine replicates $observer^{-1}$) following a random order of observation. The analysis of the replicates was realized characterizing between 50 and 90 oocytes using an optical microscope (Zeiss Axioskop 2 Plus, Carl Zeiss Microscopy LLC). The characterization was based on the morphology of the oocytes (non-spherical or spherical; Figure 2a,b) and the integrity of the chorion (intact, partially removed and totally removed; Figure 2a,b). The oocytes with spherical shape and chorion partially or totally removed were considered 'active oocytes'. The ratio of active oocytes (RAO = number active oocytes × total $oocytes^{-1}$) was calculated for each replicate.

2.3 | Fertilization in vitro

The repeatability of the protocol and the data consistency were tested by performing the fertilization in vitro experiment twice, using the same factors, but with one-month interval and employing two different stocks of adult specimens. Four females and four males were selected for each assay. The 1st assay took place in January, the females presented CL = 53.4 ± 4.7 mm, TW = 16.6 ± 4.6 g, GSI = 39 ± 8; and the males CL = 53.4 ± 7.1 mm, TW = 15.7 ± 5.6 g. The 2nd assay was realized in February, the females presented CL = 50.2 ± 4.1 mm, TW = 13.9 ± 4.5 g, GSI = 47 ± 8; and males CL = 50.8 ± 6.9 mm, TW = 17.6 ± 9.0 g.

The pH and AT treatments were selected following the higher RAO from the 'Artificial maturation' experiment. Two pH treatments plus control were tested: 8.0, 9.0 and 9.5 ± 0.1. Three AT were selected: 3:00 and 4:00 h due to their effectiveness in the 'Artificial maturation' experiment, and 1:30 h as a precaution, since longer times could be excessive in comparison with previous studies (Damen & Dictus, 1994; González-Novoa, 2014; Hodgson et al., 2007; Pabst-Fernández, 2015; Wanninger et al., 1999). Additionally, three fertilization times (FT: 1:30, 3:00 and 24:00 h) were selected to determine the time required to ensure the fertilization of the oocytes. In total 27 combined treatments (pH × AT × FT) were applied and four replicates *per* combined treatment. The activation of the oocytes was made in a single crystal beaker (600 ml) *per* pH × AT treatment, resulting a total of nine beakers. After the activation, the oocytes were washed with FSU using a 100 μm mesh. Then, they were kept in 500 ml FSU until the start of the fertilization. As replicates were

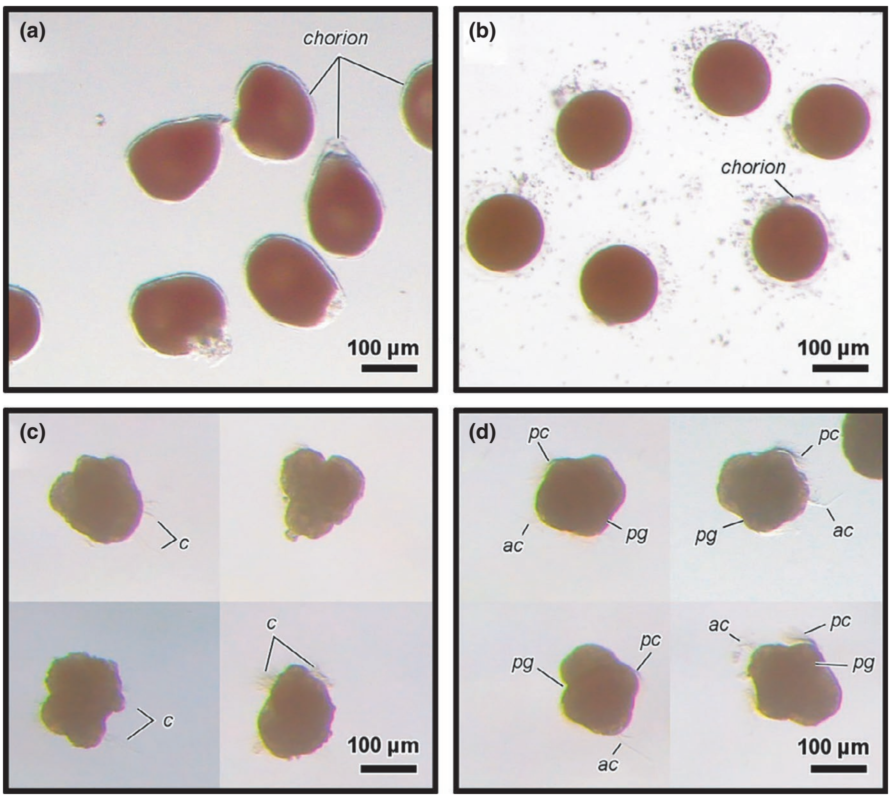

FIGURE 2 *Patella aspera*. Leica M165 C Stereo microscope. (a) Oocytes with non-spherical shape and intact chorion, minutes after extraction from the gonad. (b) Active oocytes (spherical shape and chorion partially or totally removed), after an alkalinized seawater bath at pH 9.5 and AT = 3:00 h. (c) Larvae showing abnormal morphology, 24 h post-fertilization. (d) Trocophora larvae showing normal morphology, 24 h post-fertilization. Abbreviations: ac, apical cilia; c, cilia; pc, prototrochal cilia; pg, prototrochal girdle

used 100 ml plastic cups filled with 40 ml FSU and 60 ± 8 oocytes ml^{-1}. The sperm was added for a final concentration around 10^6 sperm cells ml^{-1}. Immediately after the start of the fertilization, a small sample of oocytes *per* combined pH × AT treatment was characterized following the 'Artificial maturation' experiment. In the treatments FT = 1:30 and FT = 3:00 h, when the FT finished, the oocytes were washed and resuspended with another 40 ml FSU. In the FT = 24:00 h treatments, oocytes and sperm were kept together for 24 h. The incubation occurred at room temperature (19.5 ± 0.5°C in 1st assay; 18.0 ± 0.5°C in 2nd assay) and natural salinity (37 ± 1 g/L). The incubation lasted 24 h since the start of the fertilization.

Sampling was performed at the end of the incubation period, adding 2 ml of absolute ethanol to induce the sinking of the swimming larvae. The specimens were collected from the bottom of the plastic cup and fixed in 5.6% formaldehyde in filtered seawater taking a single sample *per* replicate. All the specimens from each sample were checked using a stereomicroscope (Leica M165 C & Leica M80, Leica Microsystems) and were categorized as oocytes (presumable non-fertilized oocytes), abnormal larvae (deformed bodies, cell clusters, presumable decomposed specimens; Figure 2c) and normal larvae (Figure 2d). The larval production (LP = number normal larvae × total specimens^{-1}), the ratio of abnormal larvae (RAL = number abnormal larvae × total specimens^{-1}) and the ratio of oocytes (RO = number oocytes × total specimens^{-1}) were calculated for each replicate.

2.4 | Statistical analysis

All statistical analyses were performed using R software version 3.6.3 (R Development Core Team, 2018). The following variables were arcsine square root transformed to meet the ANOVA assumptions: ratio of active oocytes (RAO), larval production (LP), ratio of abnormal larvae (RAL) and ratio of oocytes (RO). The homogeneity of variances was tested using the Levene's test ('car 3.0-7', Fox & Weisberg, 2019) and the normality of the residuals using the Shapiro–Wilk test, establishing a critical level (α) of 0.05 to reject the null hypothesis.

A two-way design (two-way ANOVA, type III) was used to analyse the RAO using pH and AT as factors ('car 3.0-7'), and post hoc *t* tests with Bonferroni correction to analyse the influence of the pH within each activation time ('lsmeans 2.30-0', Lenth & Lenth, 2018), being established a critical level (α) of 0.05 to reject the null hypothesis.

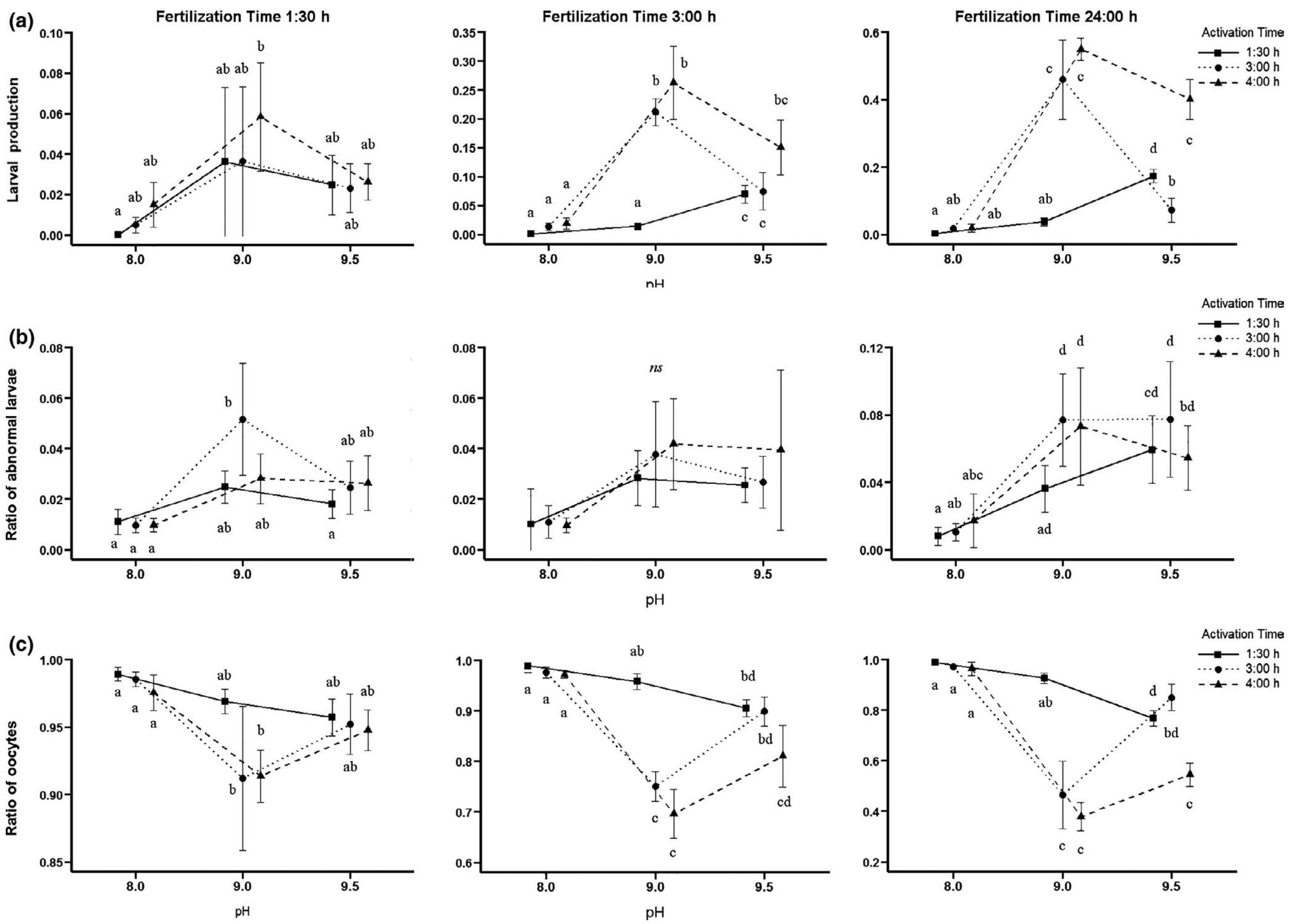

FIGURE 3 *Patella aspera*. Fertilization in vitro, 1st assay. Interaction plots within each Fertilization Time subset. (a) Larval production. (b) Ratio of abnormal larvae. (c) Ratio of oocytes. The different letters indicate significant differences among treatments ($p < 0.01$). The abbreviation *ns* indicates no significant differences among treatments

The dependent variables LP, RAL and RO were analysed by a three-way design based on robust ANOVA («t3way» function, 'WRS2 1.1-0'; Mair & Wilcox 2020) using pH, AT and FT as factors. A more conservative critical level (α) of 0.01 was selected. Then, the data sets were subdivided into FT subsets (FT = 1:30, 3:00 and 24:00 h) to compare the pH × AT combinations using post hoc t tests, but since a minority of the subsets did not met the assumptions of homogeneity and normality, a critical level (α) of 0.01 was maintained.

The relationship between LP and RAO in the 1st and 2nd assays was analysed using linear regression models: lm(LP ~ RAO, SUBSET). Three linear regression models *per* assay were performed, each one corresponding to one FT subset: 1:30, 3:00 and 24:00 h. The statistical significance was established with a critical level (α) of 0.05 to reject the null hypothesis.

3 | RESULTS

3.1 | Artificial maturation

The ratio of active oocytes (RAO) was not affected by pH, but it was significantly affected by AT. The interaction pH × AT was significant (Table S1). The RAO was lower during the first hour, without significant differences among pH treatments. From AT = 3:00 to 5:00 h, the RAO was higher at pH 9.0 and 9.5 and lower at pH 8.0, 8.5 and 10.2 (Table S2).

3.2 | Fertilization in vitro

In the 1st assay, in average 253 ± 91 specimens replicate^{-1} were characterized. The larval production (LP) was significantly affected by all the single factors (pH, AT and FT) and all their interactions (Table S3). The higher LP was observed at pH 9.0 × AT ≥ 3:00 h × FT ≥ 3:00 h, while the lower LP was observed at pH 8.0 or FT = 1:30 h (Figure 3a; Table S4). The LP increased from pH 8.0 to 9.0, then decreased from pH 9.0 to 9.5 (Figure 3a). This trend, as well the overall LP, increased with longer FT (Figure 3a).

The three-way robust ANOVA showed that pH × FT was the single interaction influencing the ratio of abnormal larvae (RAL; Table S3). The higher RAL was observed at pH ≥ 9.0 × FT = 24:00 h, while the lower RAL was observed at pH 8.0 (Figure 3b; Table S5). Apparently, RAL increases when pH > 8.0, but it was only significant at FT = 24:00 (Figure 3b).

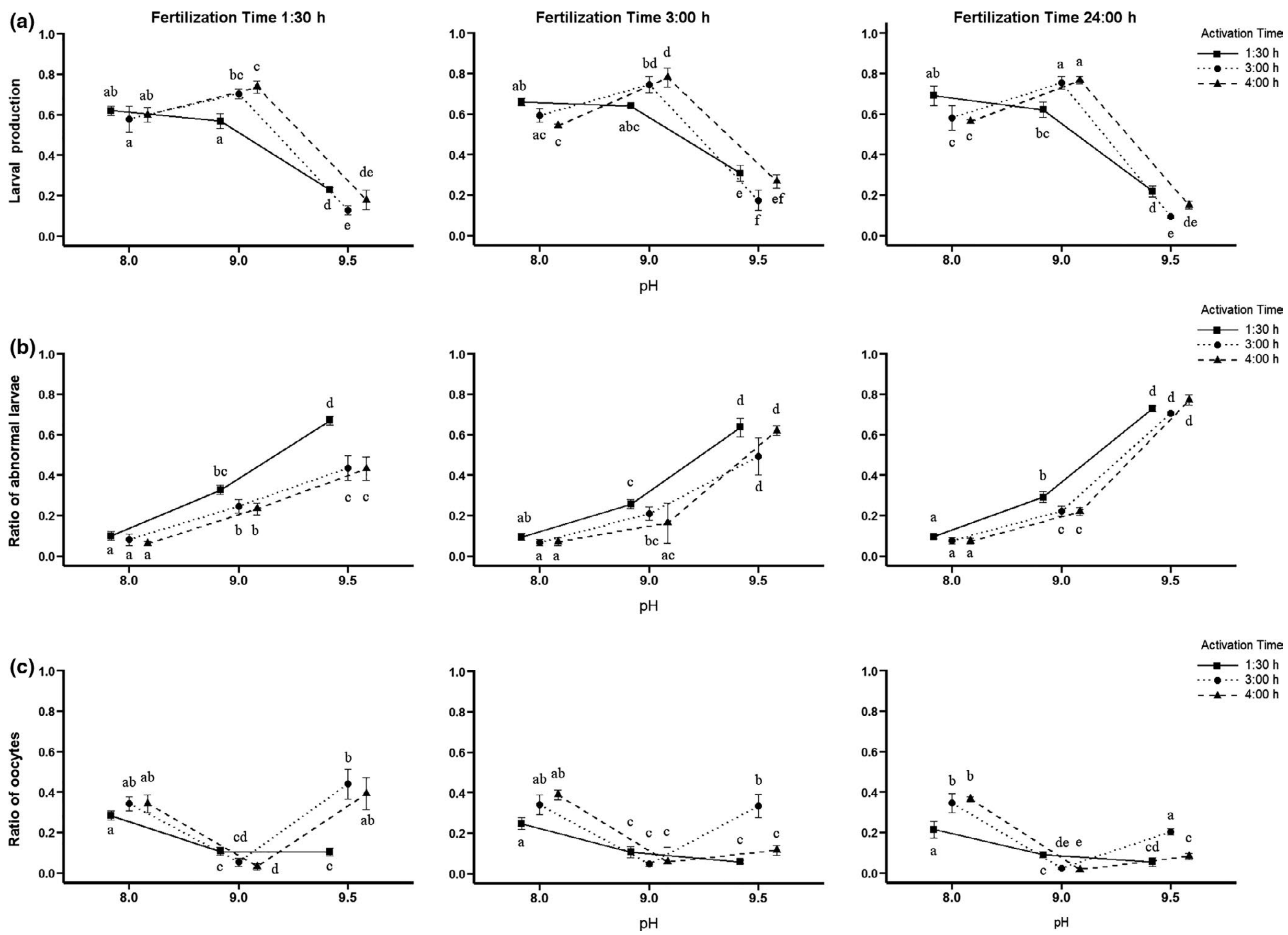

FIGURE 4 *Patella aspera*. Fertilization in vitro, 2nd assay. Interaction plots within each Fertilization Time subset. (a) Larval production. (b) Ratio of abnormal larvae. (c) Ratio of oocytes. The different letters indicate significant differences among treatments ($p < 0.01$)

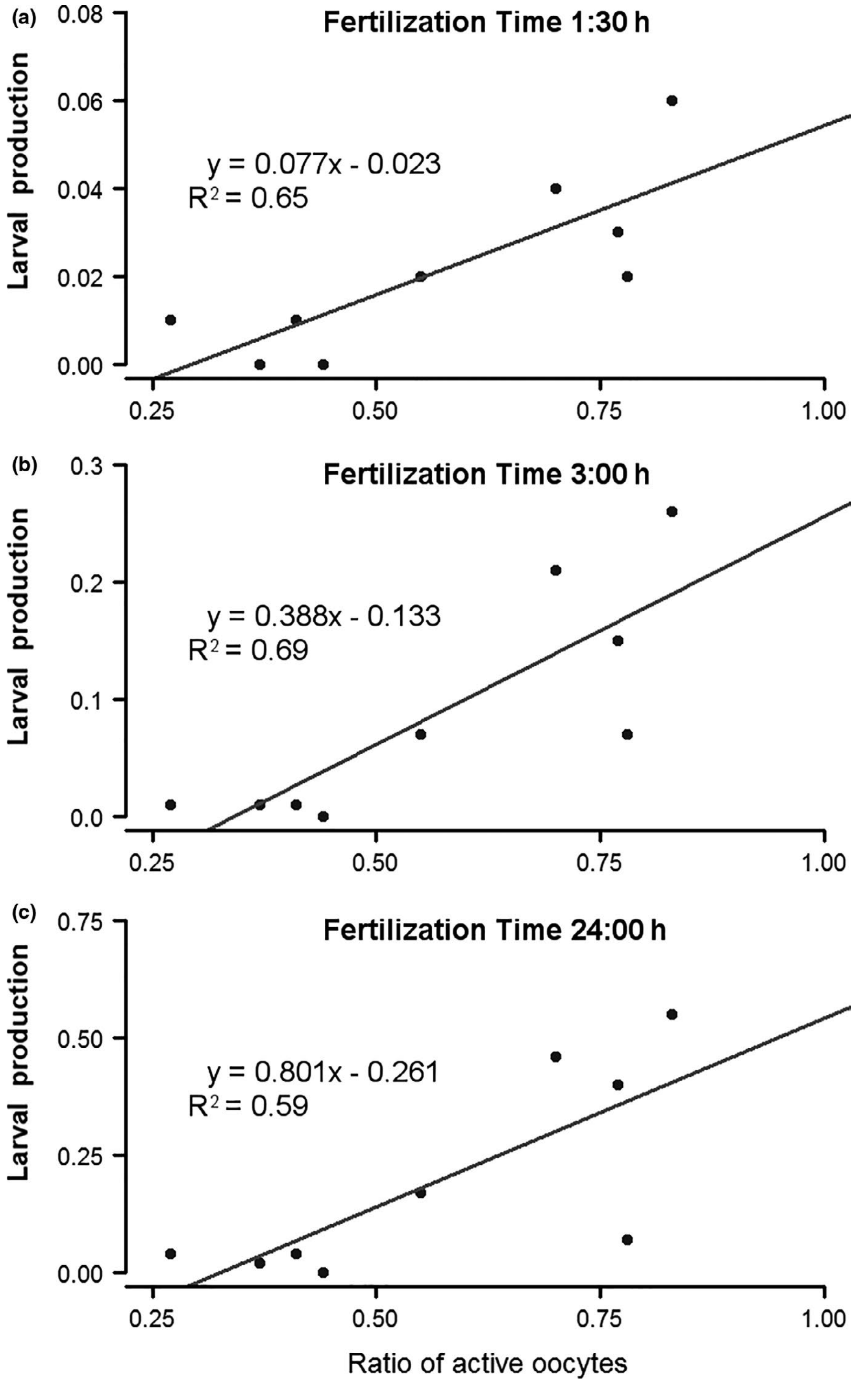

FIGURE 5 *Patella aspera*. Fertilization in vitro, 1st assay. Linear regression models between the larval production and ratio of active oocytes at each Fertilization Time. (a) Fertilization Time = 1:30 h. (b) Fertilization Time = 3:00 h. (c) Fertilization Time = 24:00 h

Similarly to LP, the ratio of oocytes (RO) was significantly affected by all the single factors and their respective interactions (Table S3). The RO was lower at pH 9.0 or 9.5 combined with AT ≥ 3:00 h × FT ≥ 3:00 h (Figure 3c; Table S6). Contrary to LP, RO decreased from pH 8.0 to 9.0 and then increased from pH 9.0 to 9.5. This trend was more pronounced with longer FT (Figure 3c).

Regarding the 2nd assay, an average of 426 ± 174 specimens replicate^{-1} was analysed. The LP was significantly affected by all the single factors (pH, AT or FT), and the interactions pH × AT and pH × FT (Table S7). The higher LP was observed at pH 9.0 × AT ≥ 3:00 h, and the lower LP occurred at pH 9.5 (Figure 4a; Table S8). The general trend of LP was to increase from pH 8.0 to 9.0 and to decrease from pH 9.0 to 9.5 (Figure 4a). This trend was more pronounced when AT ≥ 3:00 h (Figure 4a). Regarding FT, no clear trend was observed (Figure 4a).

RAL was significantly affected by all the single factors and all the interactions, except for pH × AT (Table S7). The higher RAL was observed at pH 9.5, with a peak at FT = 24:00 h, while the lower RAL

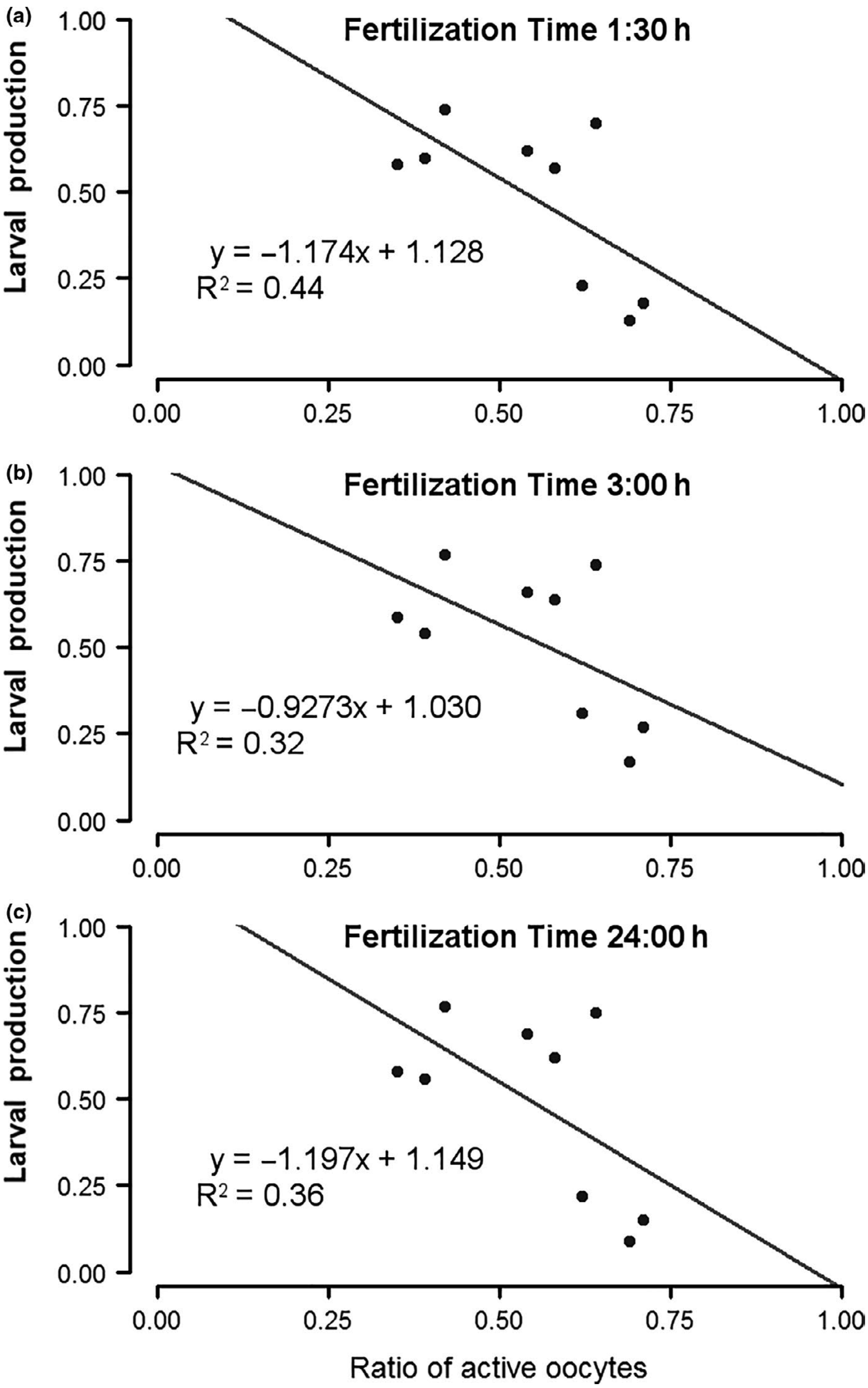

FIGURE 6 *Patella aspera*. Fertilization in vitro, 2nd assay. Linear regression models between the larval production and the ratio of active oocytes at each Fertilization Time. (a) Fertilization Time = 1:30 h. (b) Fertilization Time = 3:00 h. (c) Fertilization Time = 24:00 h

occurred at pH 8.0 (Figure 4b; Table S9). In general, RAL increased with pH, the influence of AT was not clear, and apparently, RAL at pH 9.5 increased with longer FT (Figure 4b).

All the single factors and all the interactions significantly affected RO, except interaction AT × FT (Table S7). The higher RO was observed at pH 8.0 together with pH 9.5 × AT = 3:00, while the lower RO occurred at pH 9.0 together with pH 9.5 × AT = 1:30 (Figure 4c; Table S10). The RO decreased from pH 8.0 to 9.0 and increased from pH 9.0 to 9.5, showing a less pronounced slope with longer FT (Figure 4c; Table S10).

3.3 | Relationship between RAO and LP

The results from 1st and 2nd assays showed a differential behaviour and effectiveness regarding the process of activation of the oocytes. In the 1st assay, RAO was significantly influenced by pH, AT, as well the interaction pH × AT (Table S11). The higher RAO was observed at pH 9.0 and pH 9.5 combined with AT ≥ 3:00 h, while the lower RAO occurred at pH 8.0 and AT = 1:30 h (Table S12). LP significantly increased with RAO following a linear regression model (Figure 5; Table S3).

By contrast, in the 2nd assay, no single factor or interaction showed statistical significance (Tables S11 and S12). Regarding the relationship between RAO and LP, all the regression linear models were not significant, showing low R^2 values (Table S13). The LP decreased with RAO (Figure 6).

4 | DISCUSSION

The present study adapted successfully the use of NaOH-alkalinized seawater baths to enhance the larval production in the limpet *P. aspera*. Recommended conditions are as follows: females captured in January or February with GSI between 40 and 50, alkalinised baths at pH 9.0 during 3:00 h, then the oocytes should be washed and sperm added at 10^6 sperm cells ml^{-1}. The larval production ranged from 50% to 75% after 24 h incubation at 19 ± 1°C.

All the experiments showed a significant interaction between pH and activation time influencing the larval production. This suggests that the activation is a time-dependent process in which the pH influence requires a given time to be effective. For *P. aspera*, the optimal AT was between 3 and 4 h, which is a long AT since few studies reported activation times longer than one hour (Aquino De Souza et al., 2009; Guallart et al., 2020; Pérez et al., 2019), and usually no more than 10 min (Damen & Dictus, 1994; González-Novoa, 2014; Hodgson et al., 2007; Pabst-Fernández, 2015; Pérez et al., 2019; Wanninger et al., 1999). RAO decreases at pH 10 or higher, as reported in other limpet species (Aquino De Souza et al., 2009), probably due to the degradation of the cellular membranes (Aquino De Souza et al., 2009) and the damage of the oocytes (González-Novoa, 2014).

In both 'Fertilization in vitro' assays, higher larval production occurred at pH 9.0 combined with AT = 3 and 4 h. However, treatments pH 8.0 and 9.5 showed a differential influence between assays: LP at pH 8.0 was anecdotal in the 1st assay but around 60% in the 2nd assay; while RAL at pH 9.5 was lower than 12% in the 1st assay but more than 70% in the 2nd assay. These differences may respond to biological traits: the higher ratio of *P. aspera* with gonads at the highest maturation stage occurs in February (Sousa et al., 2017), suggesting that the gonads of the 2nd assay contained more fully ripped oocytes, probably ready to be fertilized at natural conditions and explaining the higher larval production at control (pH 8.0). Simultaneously, ripped oocytes could be more vulnerable to the activation, leading to a higher ratio of abnormal larvae at a pH otherwise tolerable, explaining the higher RAL at pH 9.5 in the 2nd assay. In fact, previous studies reported that a high pH lead to increased RAL, *for example P. depressa* (González-Novoa, 2014); or reduced larval production, *for example P. vulgata* (Pabst-Fernández, 2015). RAO as an effective predictor of the LP requires further research due to the contradictory results obtained in this study. Finding predictors regarding the larval production success is a required step given their usefulness as tools to optimize future production processes.

Regarding the sperm, our results show that the sperm of *P. aspera* directly extracted from the gonad does not require any special treatment to carry out an effective fertilization, similarly to other patellids (Hodgson et al., 2007; Pabst-Fernández, 2015; Pérez et al., 2019) and differing from other molluscs, such as oysters (Demoy-Schneider et al., 2012; Ohta et al., 2007) and scallops (Faure et al., 1994; Suquet et al., 2013) that requires special treatments to undergo activation when diluted in seawater, *for example* to transit the sexual tract or variations in pH or NH_3 concentration.

Mau and Jha (2018a) evaluated the dissection and fertilization in vitro as unreasonable methods to obtain larvae for aquaculture production, since these methods were considered lengthy and intricate. However, the present study demonstrates that dissection and fertilization in vitro protocol is in fact a reliable methodology to produce *P. aspera* larvae. The major advantages of the present methodology include affordable and easy to obtain reagents, acclimated adult specimens are not a requisite, and larval production is ensured 24 h after fertilization. The results of the present study have direct practical implications for *P. aspera* larvae production for both laboratory experiments and larger scale trial culture of this limpet species. Larval production of limpets in controlled conditions is a key step for a better understanding of their reproductive and larval biology. Next steps towards this goal should be to determine the seasonal timing and specimen condition required to obtain high-quality larvae, as well optimal alkalinising agents, gamete concentrations and environmental parameters. Additionally, from an aquaculture point of view, larvae obtained by this method could be employed to study the conditions required for larval culture and settlement. Other authors also pointed the interest of limpets larvae for ecotoxicological studies (Pérez et al., 2016; Pérez et al., 2019).

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CONFLICT OF INTEREST

The authors have no conflict of interest to declare.

FUNDING INFORMATION

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AUTHOR CONTRIBUTION

Castejón, D. Experimental design, Experiment realization, Samples analysis, Statistical analyses, Drafted paper; Cañizares, J.M. Animal maintenance, Experiment realization, Samples analysis, Drafted paper review; Nogueira, N.C. Drafted paper review, Project elaboration; Andrade, C.A.P. Drafted paper review, Project elaboration, Coordination and direction.

ETHICAL APPROVAL

The authors confirm that the ethical policies of the journal, as noted on the journal's author guidelines page, have been adhered to. All applicable international, national and/or institutional guidelines for the care and use of animals were followed.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

ORCID

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SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section.

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